Institute of Computer Science
University of Augsburg

Bachelor's Thesis

Title of the Bachelor's or Master's Thesis

Name

Student ID Number:	Student ID Number
First Reviewer:	Prof. Dr. Jakob Siegfried Kottmann
Second Reviewer:	Prof. Dr. Mónica Benito
Scientific Supervisor:	
Date:	March 4, 2025

written at
Chair of Quantum Algorithms
Prof. Dr. Jakob Siegfried Kottmann
Institut of Computer Science
University of Augsburg
D–86135 Augsburg, Germany

Abstract

Describe the content of the present thesis here. Briefly explain the starting point of the thesis and outline its objectives. Then, discuss the methods used and the results achieved.

Contents

1. Introduction

This document serves as a template for your thesis. At the same time, it contains important information designed to make writing your thesis easier.

Provide a general introduction to the topic of your thesis here.

1.1. Objectives

Describe the objectives of your thesis here.

1.2. Overview

Explain how you will proceed in the following chapters to achieve the objectives defined above. Chapter 2 contains information on how to structure the content of your thesis. In Chapter 3, you will find a brief introduction to LaTeX and important packages that may be useful for your work. Chapter 4 concludes this document.

2. Notes on Content

Your thesis is a scientific work and must meet corresponding quality standards.

- Do not use colloquial expressions.
- Ensure that all statements you make are supported by appropriate arguments or literature references.
- Perform a spell check before submission. A common tool for this is `aspell`, which can also be used in editors like Emacs.

A small test to check if the compilation works.

3. Brief Introduction to LaTeX

3.1. Structure

Structure your document by using chapters (`\chapter`), sections (`\section`), and subsections (`\subsection` and `\subsubsection`). If these are not sufficient, you can also use paragraphs (`\paragraph`). In the template used here, numbering is automatically generated for the sections up to the `subsection` level, and corresponding entries are added to the table of contents. If you do not want this at certain places, use the *-form of the command.

3.2. Text Formatting

LaTeX offers various ways to structure a text. The following sections briefly show the usage of lists, tables, and images.

3.2.1. Lists

In LaTeX, there are three types of lists. The most commonly used is the `itemize` environment:

- This is the first list item.
- Another list item.
- Lists can of course also consist of longer and multiple paragraphs. LaTeX will handle the correct sentence structure of the list text just like normal text.

Sometimes, you may want to number the list items. For this, the `enumerate` environment is used:

1. Individual list items are also set here using `\item`.
2. The numbering is automatically generated by LaTeX.
3. Lists can also be nested:
    - Here with an unordered list.

If you want to explain several terms, the `description` environment is suitable:

Usage Descriptions are set like any other list. But remember to provide the term being described as an option to the `\item`.

Text Formatting You can format individual parts of the text using `\textXX` to make them **bold**, *italic*, or SMALL CAPS to emphasize them. For simple emphasis, it is better to use the `\emph` command, which can also be nested.

Document Class The first command of a LaTeXdocument specifies the document class (template used). For example, this document uses the class `sikthesis` (`\documentclass{sikthesis}`). You can provide an optional argument `[draft]` to the document class to generate a draft version of the document. Images will then only be displayed as empty frames, and LaTeX will mark places at the margins where there are issues with text overflow or line breaks. These issues can usually be resolved by slightly rephrasing the text.

3.2.2. Tables

A table in LaTeXis created using the `tabular` environment and embedded within `table` environments. LaTeXplaces these tables *floating*. Such tables should be provided with a `\caption` and, if necessary, referenced in the text (see Table 3.1). If the `\caption` text is very long, it is recommended to provide a short title for the table of contents as an optional argument to the macro:

`\caption[Short title for table of contents]{Full caption}`.

Table 3.1.: Table Title

a	b	c
1	2	333333333
1	2	3333333333
1	2	3333333333
1	2	3333333333

3.2.3. Figures

Like tables, figures should be placed in a floating manner, so that LaTeXhandles their placement. Figure 3.1 shows an example of such an image. The macro `\missingfigure{...}` from the `todonotes` package is used as a placeholder for a graphic.

To create graphics, it is recommended to use the Ti*k*Zpackage. This allows you to describe the graphic directly in the TeXcode. This ensures that labels in the graphic match the font used in the rest of the document. An example of such a graphic can

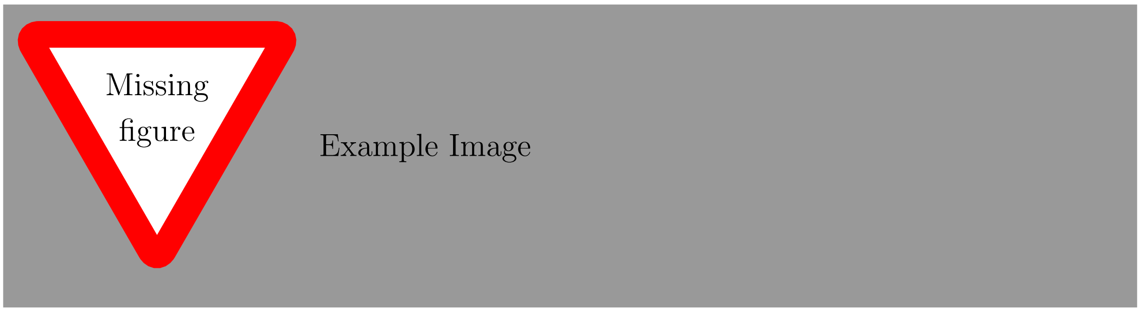

Figure 3.1.: Example of an image, 60% text width, though the width can also be specified absolutely, e.g., 6cm

be found in Figures 3.2 and 3.3. These figures also show how you can place two graphics side by side using `minipage` environments.

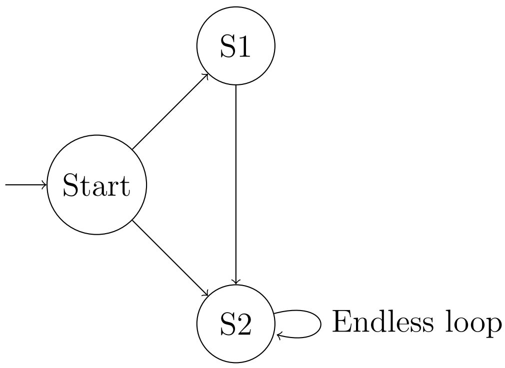

Figure 3.2.: Ti*k*ZGraphic 1

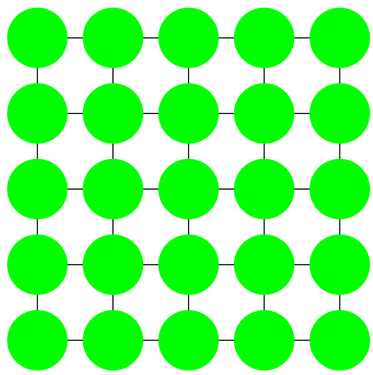

Figure 3.3.: Ti*k*ZGraphic 2

To create plots, you can use the `pgfplots` package, which is based on Ti*k*Zand PGF. Examples can be found in Figures 3.4 and 3.5.

3.3. Set of Mathematical Symbols

Mathematical formulas in the text are marked using the \$-sign, e.g. $E = mc^2$. Displayed formulas are set as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

If you want to number the formulas, use the `equation` environment:

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) dt \tag{3.1}$$

Now the formula can be referenced in the text using `\eqref`: Equation (3.1).

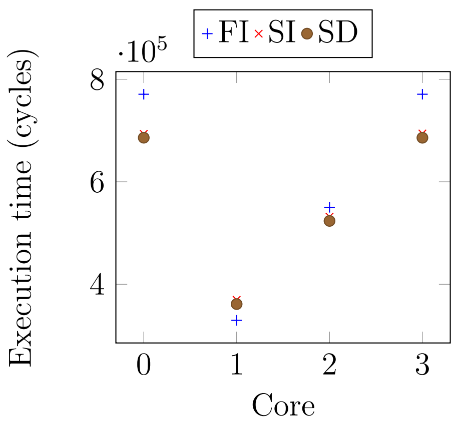

Figure 3.4.: pgf Plot 1

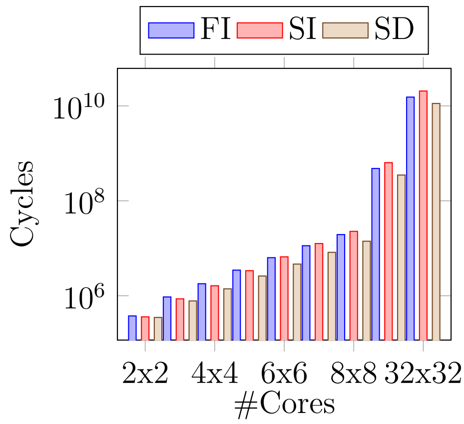

Figure 3.5.: pgf Plot 2

Note that LaTeX treats each symbol as its own variable in mathematical mode. For instance, the string $size$ will be interpreted as the product of the variables s, i, z, and e (the multiplication sign is usually omitted). If you want to use a variable with a longer name, you can set it using `\mathit{...}`, for example, size. If you look closely, you'll notice that the spacing between the characters is handled differently in this case.

3.4. External References

References are managed using BibTeX. The references are stored in a separate file (e.g., `thesis.bib`) and given abbreviations. These can then be referenced in the text. Note that you will need to compile your document twice. Between the two `latex`- or `pdflatex`-runs, you need to run `bibtex`. For more information on this and general LaTeX usage, refer to [2] and the internet.

3.5. Helpers

This template includes several packages that should help you with working in LaTeXand preparing your thesis:

acro allows for easy usage of acronyms without having to manually track where they have been used. Acronyms can be defined in the preamble of the document using `\DeclareAcronym{id}{short=...,long=...}`. In the text, the acronym is then used with `\ac{id}`. The first time it is used, the long form along with the abbreviation is displayed (e.g., „operating system (OS)“), and subsequent uses will only show the abbreviation (e.g., „OS“). A list of all available acronyms can be generated using `\printacronyms`.

todonotes is useful for ongoing work on the document. It allows you to create margin notes that contain reminders for later revisions. For example, . Additionally, with this package, placeholder graphics such as the one in Figure 3.1 can be created. Using `\todototoc` will generate a list of all existing todo notes.

Example of a todo note

4. Summary and Outlook

List of Figures

List of Tables

Bibliography

[1] Heise online.

[2] H. Kopka. *LaTeX, Band 1.* Addison-Wesley Verlag, 2000.

[3] C. L. Liu and J. W. Layland. Scheduling algorithms for multiprogramming in a hard-real-time environment. *Journal of the ACM*, 20(1):46–61, Jan. 1973.

A. Appendix Chapter

If you want to add information to the thesis that is only marginally relevant for understanding your work, create an appendix for this purpose. Internet reference [1], reference with multiple authors [3]

A.1. foo

A.2. bar

B. Second Appendix Chapter